Categorical Time-Series Analysis

General Notation

(X_t) is a (categorical (ordinal/ nominal)) *stationary* process , if

1. $E(X_t) = \mu_X \quad \forall t$
2. $E(X_t^2) < \infty \quad \forall t$
3. $Cov(X_t, X_{t+h}) = \gamma(h) \quad \forall t,\ h$

with range $S = \{s_0, \ldots, s_m\}$ $m \in \mathbb{N}$. While the order of magnitude of the elements in S represents the natural of the elements in the ordinal case, it is arbitrary in the case of nominal S and only used for readability and consistency in notation.

wo kommt die Notation her?

Analytical tools for ordinal time series are using the marginal CDF ($\boldsymbol{f}$) while tools for nominal data are based on the marginal PMF ($\boldsymbol{p}$): warum unterschiedlich?

$$\boldsymbol{f} := F(X_{t\ (ord.)}) = (f_0,\ \ldots, f_{m-1})^T = (P(X_{t\ (ord.)} \leq s_0),\ \ldots, P(X_{t\ (ord.)} \leq s_{m-1}))^T$$

oben geht es bis m

$$\boldsymbol{p} := f(X_{t\ (nom.)}) = (p_0,\ \ldots, p_m)^T = (P(X_{t\ (nom.)} = s_0),\ \ldots, P(X_{t\ (nom.)} = s_m))^T$$

And bivariate (cumulative) probabilities with time lag (h) are accordingly:

$$f_{ij}(h) = P(X_t \leq s_i, X_{t+h} \leq s_j) \quad j, i \cancel{\leq} m \quad \text{in } \{0, 1, \ldots, m\}$$
$$p_{ij}(h) = P(X_t = s_i, X_{t+h} = s_j) \quad j, i \leq m$$

Location

For ordinal data, the measure of location if given by the median and for ordinal its the mode

$$Med(X_{t\ (ord.)}) = min(s_i \in S : F(s_i) \geq \frac{1}{2})$$

bitte mit \ arbeiten für min, max

$$Mod(X_{t\ (nom.)}) = s_i \in S : P(X_t = s_i) = argmax(\boldsymbol{p})$$

saubere Notation diskutieren

Measures of Dispersion

Dispersion of categorical variables is commonly quantified via the *Index of Ordinal Variation* (IOV) for ordinal data and the *Index of Qualitative Variation* (IQV) for nominal data. Both measures take values in the interval $[0, 1]$, where 0 indicates no dispersion (degenerate one–point distribution) and 1 indicates maximal dispersion. Warum? Nachweis oder Referenz angeben

For ordinal variables, dispersion is defined in terms of the CDF vector $\mathbf{f}$:

$$\text{IOV} = \frac{4}{m} \sum_{i=0}^{m-1} f_i(1 - f_i), \text{ where } \cancel{f_i = P(X_t \leq s_i)}$$

steht schon oben

For nominal variables, dispersion is defined in terms of the PMF $\mathbf{p}$:

$$\text{IQV} = \frac{m+1}{m}\left(1 - \sum_{i=0}^{m} p_i^2\right), \qquad \cancel{p_i = P(X_t = s_i)}.$$

Minimal Dispersion

Both measures attain the value 0 if and only if the distribution is degenerate, that is, if all probability mass is concentrated in a single category. → Ok → bitte nachrechnen

$\Longleftrightarrow$

Ordinal case. A one–point distribution is characterized by a CDF vector $\mathbf{f}$ of the form

$$\mathbf{f} \in \left\{ \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\}.$$

→ hier wird nur ⇐ gezeigt

gut

In each case, at every index i either $f_i = 0$ or $f_i = 1$, hence

$$f_i(1 - f_i) = 0 \qquad \forall i.$$

Therefore,

$$\text{IOV} = \frac{4}{m} \sum_{i=0}^{m-1} f_i(1 - f_i) = 0.$$

Nominal case. A degenerate distribution has a PMF of the form

$$\mathbf{p} \in \left\{ \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\}.$$

Thus,

$$\sum_{i=0}^{m} p_i^2 = m \cdot 0 + 1 \cdot 1 = 1,$$

and therefore

$$\text{IQV} = \frac{m+1}{m}(1 - 1) = 0.$$

Maximal Dispersion

Ordinal case. Dispersion of an ordinal variable is maximal when half of the probability mass lies in the lowest category and half in the highest category; all intermediate categories receive probability zero. This results in a CDF vector

$$\mathbf{f} = \left(\tfrac{1}{2}, \tfrac{1}{2}, \dots, \tfrac{1}{2}\right)^\top.$$

gut

Then

$$f_i(1 - f_i) = \frac{1}{2}\left(1 - \frac{1}{2}\right) = \frac{1}{4} \qquad \forall i,$$

and thus

$$\text{IOV} = \frac{4}{m} \cdot \sum_{i=0}^{m-1} \frac{1}{4} = \frac{4}{m} \cdot \frac{m}{4} = 1.$$

warum geht es nicht größer?

→ wenn Nachweis in Anlehnung an Lit., dann Referenz angeben

Nominal case. Maximal dispersion is obtained when probability mass is distributed uniformly across all $m+1$ categories:

$$p_i = \frac{1}{m+1}, \qquad i = 0, \ldots, m.$$

Then

$$\sum_{i=0}^{m} p_i^2 = (m+1)\frac{1}{(m+1)^2} = \frac{1}{m+1},$$

gl. Frage

so that

$$\mathrm{IQV} = \frac{m+1}{m}\left(1 - \frac{1}{m+1}\right) = \frac{m+1}{m} \cdot \frac{m}{m+1} = 1.$$

Samples Measures

For the estimation of the (cumulative) probabailities based on samples we use the (cumulative) relative frequencies, that can be be best expressed if switching from (X_t) to its binarization ($(\boldsymbol{Z_t})$ for ordinal variables, $(\boldsymbol{Y}_t)$ for nominal):

$$\boldsymbol{Z}_t = (Z_{t,0}, \ldots, Z_{t,m-1})^T, Z_{t,i} = I_{\{X_t \leq s_i\}} \qquad \boldsymbol{Y}_t = (Y_{t,0}, \ldots, Y_{t,m})^T, Y_{t,i} = I_{\{X_t = s_i\}}$$

Now we can estimate $\boldsymbol{p}/\boldsymbol{f}$ by taking the mean about the binarization:

$$\hat{\boldsymbol{f}} = \frac{1}{n}\sum_{t=1}^{n} \boldsymbol{Z}_t, \quad \hat{\boldsymbol{p}} = \frac{1}{n}\sum_{t=1}^{n} \boldsymbol{Y}_t$$

? To get the asymptotics of the different sample measures, we first need a CLT on $\hat{\boldsymbol{f}}/\hat{\boldsymbol{p}}$,0 As our data are not (necesarilly) indepedent, the CLT only holds for a stationary process that fullfills a strong mixing condition (e.g. alpha mixing; Bradley 2005). The stationarity of $\boldsymbol{Z_t}/\boldsymbol{Y_t}$ follows from the stationarity of X_t, as the transforming function is deterministic. !? → das ist nicht richtig für den von Ihnen genutzten

A sequence is said to be α-mixing, if:

For any event $A \in \sigma(X_s : s \leq t)$ and event $B \in \sigma(X_s : s \geq t+h)$ the α-coefficient
$\alpha(h) = \sup |P(A \cup B) - P(A)P(B)|$ goes to 0 for $h \to \infty$ nein

If α-mixing is satisfied it holds that:

$$\sqrt{n}(\overbrace{\bar{\boldsymbol{Z}}_t}^{=\hat{\boldsymbol{f}}} - \boldsymbol{f}) \xrightarrow{d} \mathcal{N}(0, \boldsymbol{\Sigma}), \qquad \text{with } \boldsymbol{\Sigma} = (\sigma_{i,j})_{i,j=0,\ \ldots, m-1}$$

$$\text{where } \sigma_{i,j} = Cov(Z_{0,i}, Z_{0,j}) + \sum_{h=1}^{\infty} \left(Cov(Z_{0,i}, Z_{h,j}) + Cov(Z_{h,i}, Z_{0,j})\right)$$

$$\text{and } Cov(Z_{t,i}, Z_{s,j}) = \begin{cases} f_{ij}(t-s) - f_i f_j & \text{if } t \geq s \\ f_{ji}(s-t) - f_i f_j & \text{if } s \geq t \end{cases}$$

sauber als Theorem formulieren

To now derive the asymptotics of the dispersion measures (scalar functions) we can apply the delta method (i.e. Theorem on 5.25 on Asymptotic Distribution of $g(\boldsymbol{X}_n)$ in the formulary).